

Supriya Pillai

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EDUCATION

University of California, Berkeley	Berkeley, CA
<u>Postgrad Program - Machine Learning and Artificial Intelligence</u>	<u>2022</u>
Griffith University	Queensland, Australia
<u>Master of Information Technology - Data Science & Management</u>	<u>2019</u>

PROFESSIONAL EXPERIENCE

Radical AI	San Francisco, CA
<u>AI/ML Engineer</u>	<u>Mar 2024 – Present</u>

- Built and deployed LLM models using GPT and Gemini for recommender systems and personalized intent engines, achieving a 30% increase in user engagement and a 20% improvement in generative output quality through innovative model architectures.
- Architected scalable **NLP pipelines on Google Cloud Vertex AI**, enhancing understanding and optimizing ML infrastructure for efficient **multi-model data** ingestion.
- Implemented **CUDA-accelerated deep learning models**, leveraging GPU parallelism to achieve a 5x speedup in training and inference times for large-scale NLP tasks.
- **Fine-tuned diffusion models** using **DreamBooth and LoRA** for text-to-image generation, improving visual content quality.
- Led cross-functional collaboration to define and solve complex business problems, translating stakeholder needs into technical solutions.

Fellowship AI	San Francisco, CA
<u>Machine learning Engineer- Fellow</u>	<u>Apr 2022 – Aug 2022</u>

- Engineered a conversational AI tool using Hugging Face Transformers for **product recommendations**, leveraging NLP libraries, TensorFlow, Spark ML to process 10,000+ user queries. **Optimized retrieval** through query intent analysis, query transformation, and hybrid search techniques.
- Integrated a chatbot with the Rasa framework for advanced dialogue management, achieving a 98% accuracy rate in identifying fashion attributes.
- Led data extraction, formatting, and validation processes to ensure high-quality datasets, enhancing business workflows by integrating ML components and increasing efficiency by 20%.

Neural Box AI	Brisbane, Australia
<u>Co-Founder – Data Scientist</u>	<u>Jun 2020 – Mar 2021</u>

- Designed deep learning and computer vision algorithms for object detection and image segmentation using **Fastai and OpenCV**, leveraging satellite and drone imagery for early bushfire detection and predictive analytics for supply chain management.
- Incorporated privacy-preserving ML techniques including Differential Privacy and Federated Learning.

Petra Inc.	Brisbane, Australia
<u>Product Data Scientist</u>	<u>Aug 2019 – Mar 2021</u>

- Utilized a robust **machine learning infrastructure** with **Python and PyTorch** to implement classification and regression techniques for **time series forecasting and anomaly detection** on large-scale structured and unstructured data using **Hadoop and SQL**. Deployed models on **AWS via Docker**, reducing total processing time by 20%.
- Employed Tableau and Power BI to design dashboards for complex **data visualization**, enhancing operational efficiency by 20% through data analysis, with a specific focus on **predictive and data analytics** and **optimization**.
- Led cross-functional AI/ML development initiatives, including experiment design and causal inference (non-A/B and RCT experiments), achieving a 10% reduction in downtime and substantial cost savings.

Griffith University Prince Charles Hospital (ICET Lab)	Brisbane, Australia
<u>Data Science Intern</u>	<u>Feb 2019 – May 2019</u>

- Formulated and refined search algorithms for 10+ researchers **using NLP and ML techniques**, resulting in a 25% increase in search relevance and a 30% increase in user engagement. Implemented a **feature engineering and ETL pipeline** to effectively manage and store user profiles.
- Engaged with 5+ cross-functional partners to drive projects forward, applying analytical and innovative thinking to learn and adapt to emerging technologies and tools. Conducted statistical analysis to enhance user segmentation and improve overall user experience.

TECHNICAL SKILLS

Python, SQL, C++, JavaScript, Scala, TensorRT, Large Language models, Prompt Engineering, GPU optimization, Big Data, Statistical Analysis, Machine Learning, Deep Learning, Azure ML, Scikit-learn, NumPy, Pandas, Data Visualization (Tableau, Power BI), AWS (EC2, Sagemaker), Amazon Bedrock, Google Cloud, Docker, Kubernetes, Model Deployment.

PROJECT EXPERIENCE

Stanford LLM x LAW Hackathon (Codex)

April 2024

- Built an **AI-powered RAG immigration assistant** utilizing GROQ's LLM and a custom knowledge base to facilitate intuitive human-computer interaction, successfully understanding user queries, maintaining conversation context, and providing tailored responses for enhanced user experience and accessibility to immigration information.

Glow Goal AI

April 2024

- Designed an innovative health journal app utilizing **Langchain, OpenAI API, Pinecone (Vector Database)** and advanced NLP techniques to provide personalized insights and support, while integrating features allowing users to add Instagram health influencers to the app.
- Created an intuitive user interface with **Streamlit** and integrated Langchain memory for contextually relevant AI-powered conversations.

Melody Mistral

March 2024

- Developed an AI-powered music app utilizing Mistral API, Spotify API, and Weather API to generate personalized music playlists based on real-time weather data, user location, and local language preferences.

AWARDS

3rd Place, Scale AI Generative AI Hackathon – GitHub HQ

Finalist, Flatten the Curve Hackathon – Australian Computing Society

Honorable Mention from Dr Junling Hu – Lumiata COVID-19 Global AI Hackathon

Ranked 2nd of 3000+ Participants, Object Detection & Headcount Computer Vision Challenge – Analytics Vidhya

Selected for Australia Defense pathway - ANZDF

Sponsored Attendee – Young women leader in AI

Nominated for Women in Tech Awards - Australia